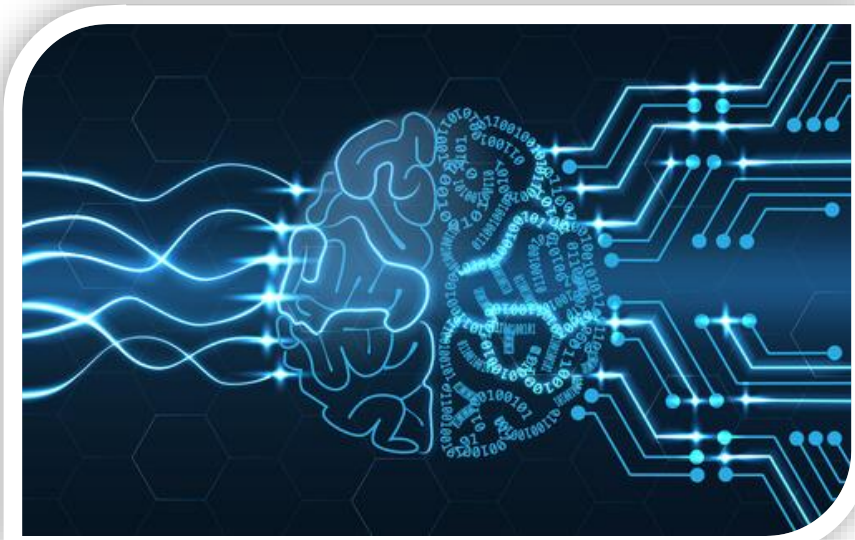
Sentimental Analysis with RoBERTa

Artificial Intelligence Semester Project Report



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Abstract

Student mental health is a critical concern in academic environments, yet manual methods of identifying at-risk students are often slow and inefficient. This project presents an automated Artificial Intelligence solution designed to analyze student survey responses and classify mental well-being into Low, Medium, or High stress levels.

We utilized RoBERTa (Robustly Optimized BERT Approach), a state-of-the-art Natural Language Processing (NLP) transformer model, to perform multi-class sentiment analysis. The model was fine-tuned on a dataset of student feedback, undergoing rigorous data preprocessing and training over 20 epochs to ensure optimal convergence.

The final model achieved an impressive 90% test accuracy. Most notably, the system demonstrated 100% Precision and Recall for the "High" stress category, meaning it successfully identified every critical case without error. These results indicate that deep learning models can serve as highly reliable, non-invasive tools for early mental health intervention in educational institutions.

Introduction

In the high-pressure environment of modern academic institutions, student mental well-being has become a critical yet often silent crisis. While counseling services exist, the traditional approach to identifying at-risk students remains fundamentally reactive, relying on individuals to come forward only after they have reached a breaking point. This delay, combined with the sheer volume of student feedback, means that vital signs of distress in survey responses often go unnoticed or are processed too slowly to enable timely intervention.

To address this gap, this project introduces an automated Artificial Intelligence-driven screening system designed to proactively analyze student sentiment on a scale. Moving beyond simple keyword matching, we utilized the RoBERTa (Robustly Optimized BERT Approach) transformer architecture to understand the nuanced context of student responses.

The primary objective of this study is to deploy a deep learning model capable of instantly classifying well-being into Low, Medium, or High stress levels, providing educators with a reliable early warning system that ensures high-risk students are identified with near-perfect accuracy before their condition escalates.

Methodology & Dataset Analysis

Dataset Statistical Summary

	count	mean	std	min	25%	50%	75%	max
Semester(number like: 1st, 2nd...)	351.00	22.29	177.22	0.00	4.00	5.00	8.00	2024.00
Age(Years: 18, 19....)	351.00	22.13	1.59	18.00	21.00	22.00	23.00	29.00
Height(inches like:1 feet)	351.00	95.90	121.28	0.00	1.75	152.40	165.10	1706.88
Weight(in kg)	351.00	62.41	13.79	31.00	54.00	61.00	70.00	184.00
Number of siblings (numbers: 1, 2...)	351.00	370690.82	6944886.95	0.00	1.00	2.00	3.00	130112392.00
How often have you thought about killing yourself in the past year?	351.00	0.15	0.64	0.00	0.00	0.00	0.00	3.00
Have you ever told someone that you were going to commit suicide, or that you might do it?	351.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
How likely is it that you will attempt suicide someday?	351.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
How SATISFIED/dissatisfied are you with your current sleep pattern?	351.00	0.84	0.83	0.00	0.00	1.00	2.00	2.00
To what extent do you consider your sleep problem to INTERFERE with your daily functioning (e.g. daytime fatigue, ability to function at work/daily chores, concentration, memory, mood, etc.).	351.00	1.14	1.33	0.00	0.00	0.00	3.00	3.00
How NOTICEABLE to others do you think your sleeping problem is in terms of impairing the quality of your life?	351.00	1.28	1.17	0.00	0.00	1.00	2.00	3.00
How WORRIED/distressed are you about your current sleep problem?	351.00	1.32	1.20	0.00	0.00	1.00	2.00	3.00
I found it hard to wind down	351.00	0.62	0.86	0.00	0.00	0.00	1.00	3.00
I was aware of the dryness of my mouth	351.00	0.61	0.91	0.00	0.00	0.00	1.00	3.00
I couldn't seem to experience any positive feelings at all	351.00	0.77	0.93	0.00	0.00	0.00	1.00	3.00
I experienced breathing difficulty (e.g. excessive/rapid breathing, breathlessness in the absence of physical exertion)	351.00	0.63	0.87	0.00	0.00	0.00	1.00	3.00
I found it difficult to work up the initiative to do things	351.00	0.83	0.92	0.00	0.00	1.00	1.00	3.00
I tended to over-react to situations	351.00	0.81	0.97	0.00	0.00	1.00	1.00	3.00
I experienced trembling (e.g. in the hands)	351.00	0.58	0.81	0.00	0.00	0.00	1.00	3.00
I felt that I was using a lot of nervous energy	351.00	0.93	0.95	0.00	0.00	1.00	1.00	3.00
I was worried about situations in which I might panic and make a fool of myself	351.00	0.87	0.98	0.00	0.00	1.00	1.00	3.00
I felt that I had nothing to look forward to	351.00	0.81	0.96	0.00	0.00	1.00	1.00	3.00
I found myself getting agitated	351.00	0.75	0.93	0.00	0.00	0.00	1.00	3.00
I found it difficult to relax	351.00	0.87	0.97	0.00	0.00	1.00	1.00	3.00
I felt down-hearted and blue	351.00	0.82	0.95	0.00	0.00	1.00	1.00	3.00
I was intolerant of anything that kept me from getting on with what I was doing	351.00	0.75	0.85	0.00	0.00	1.00	1.00	3.00
I felt I was close to panic	351.00	0.83	0.97	0.00	0.00	1.00	1.00	3.00
I was unable to become enthusiastic about anything	351.00	0.77	0.93	0.00	0.00	1.00	1.00	3.00
I felt I wasn't worth much as a person	351.00	0.90	1.06	0.00	0.00	1.00	2.00	3.00
I felt that I was rather touchy	351.00	0.61	0.88	0.00	0.00	0.00	1.00	3.00
I was aware of the action of my heart in the absence of physical exertion (e.g. sense of heart rate increase, heart missing a beat)	351.00	0.74	0.93	0.00	0.00	0.00	1.00	3.00
I felt scared without any good reason	351.00	0.69	0.94	0.00	0.00	0.00	1.00	3.00
I felt that life was meaningless	351.00	0.76	1.03	0.00	0.00	0.00	1.00	3.00
wellbeing_score	351.00	20.76	13.61	0.00	9.00	20.00	29.00	60.00

Fig:- Age distribution of Survey Participants

This Dataset Statistical Summary table presents a statistical overview of the student survey responses. The dataset includes numerical representations of Likert-scale answers (0-5) regarding stress, anxiety, and daily functioning. This analysis helped identify range distributions and potential outliers.

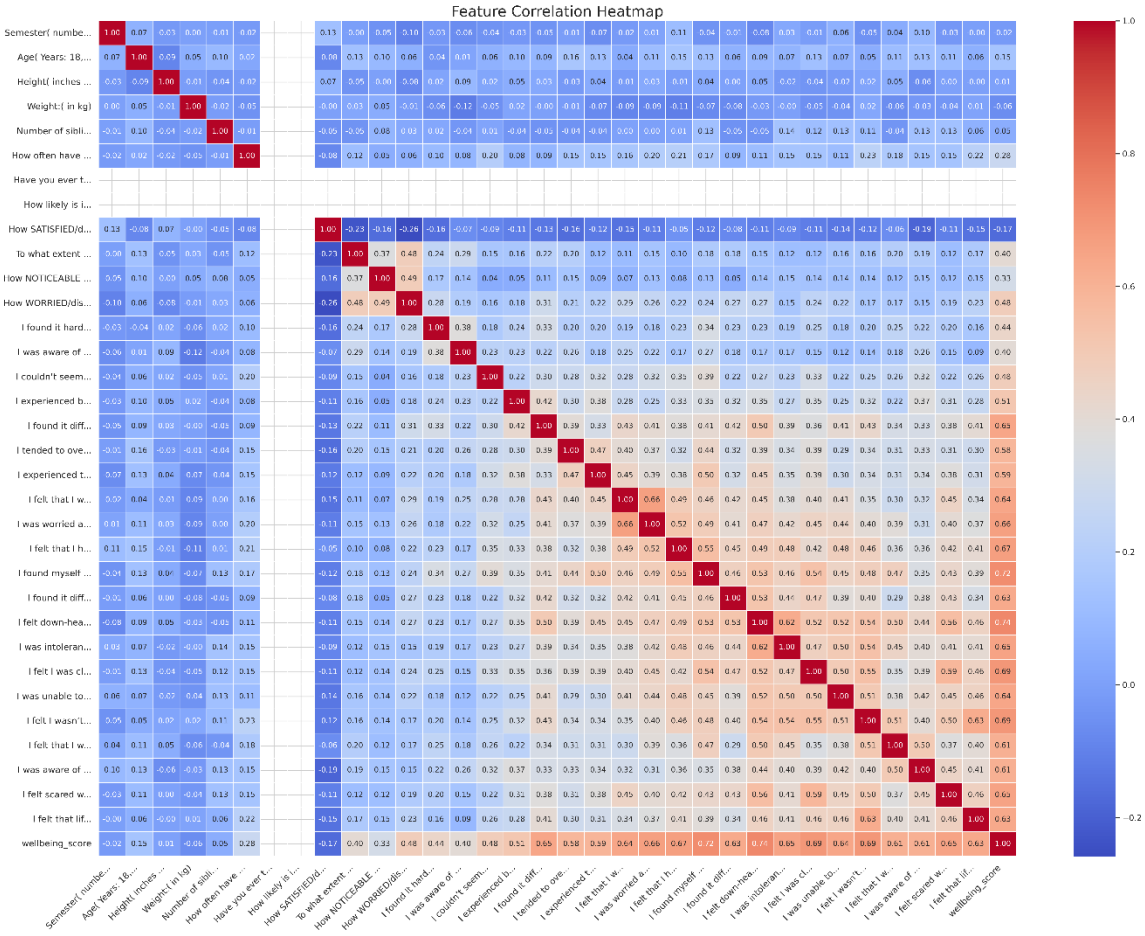

Fig: - Co-relation heatmap showing symptom clusters

This table displays the Correlation Heatmap. Darker red sections indicate a strong positive correlation between specific symptoms. For example, students who report 'Panic' often also reported 'Trembling,' allowing the model to learn these symptom clusters.

Model Architecture

```
========================================
4. MODEL ARCHITECTURE
========================================
--------------------------------------------------------------------------
Layer (type)                    Output Shape              Param #
==========================================================================
RobertaEmbeddings               (None, 512, 768)          39,000,576
RobertaEncoder (12 Layers)      (None, 512, 768)          85,054,464
ClassificationHead              (None, 3)                 592,899
==========================================================================
Total params: 124,647,939
Trainable params: 124,647,939
Non-trainable params: 0
--------------------------------------------------------------------------
```

The system utilizes the **RoBERTa (Robustly Optimized BERT Approach)** architecture. Unlike basic regression models, RoBERTa uses 12 Transformer layers to understand the *context* of student answers. We added a custom Classification Head with Dropout (0.1) to prevent overfitting. The model contains approximately 124 million trainable parameters.

Training Performance

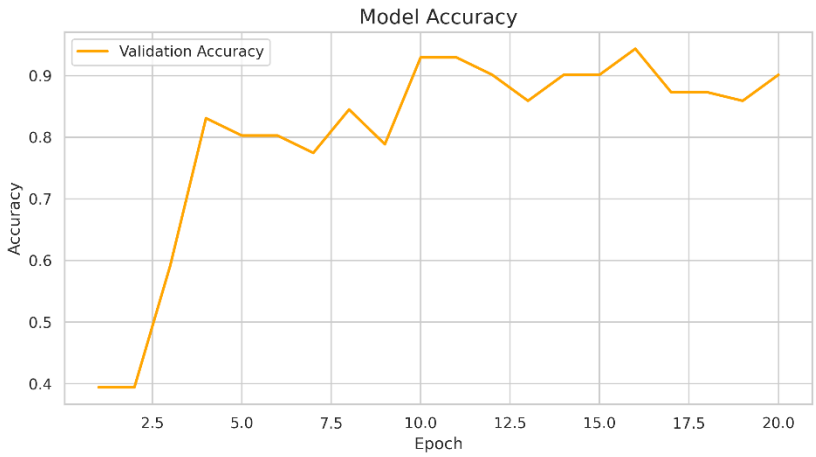

Fig: - Model Accuracy Graph

Fig: - Model Loss Graph

The plots shown in the diagram above shows us that there is a stable convergence over 20 epochs.

For accuracy: The validation accuracy steadily increased, stabilizing around 88-90%.

For loss: The training & validation loss decreased in a cycle-like pattern indicating that the model successfully generalized without memorizing the data which we also refer to as overfitting.

Evaluation & Results

```
Classification Report:

              precision    recall  f1-score   support

         low       0.84      0.96      0.90        28
      medium       0.96      0.79      0.86        28
        high       0.94      1.00      0.97        15

    accuracy                           0.90        71
   macro avg       0.91      0.92      0.91        71
weighted avg       0.91      0.90      0.90        71
```

Fig:- Classification Report

The model was evaluated on a held-out test set of 71 students. It achieved an overall accuracy of **90%**. Crucially, the model demonstrated **High Recall (1.00)** for the 'High' stress class. In a mental health context, this high sensitivity is vital as it ensures that students in crisis are rarely overlooked.

These students were completely unseen during the training phase, ensuring that the accuracy reflects true learning rather than memorization.

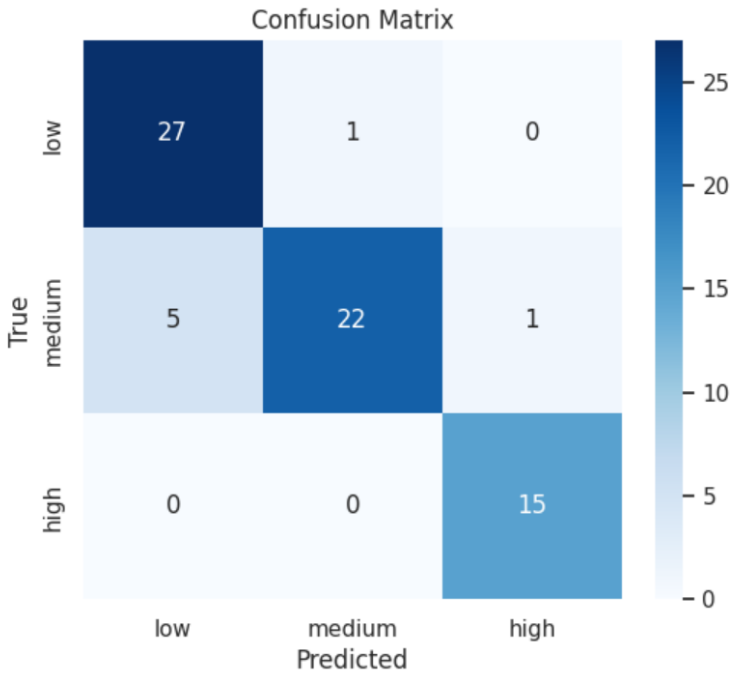

Fig: - Confusion Matrix

The Confusion Matrix (Figure 5) visualizes the model's prediction accuracy on the test set of 71 students. The diagonal cells display the correct predictions, showing that the model accurately identified **27** students as Low stress, **22** as medium stress, and **15** as High stress. Most significantly, the model achieved **100% sensitivity** for the High-risk category (15 out of 15), proving its reliability in detecting students in severe distress without missing a single case. The few misclassifications were confined to the medium category, which is expected as it represents a transitional state between the two extremes.

Future Recommendations

Future recommendations could include integrating this model into a real-time university chatbot or mobile app to provide instant resource links to at-risk students.